Machine Learning — Practice Set B Solutions

Comprehensive MTE Exam Guide • Model Answers & Step-by-Step Derivations

SECTION 1: UNIT & CONCEPTUAL QUESTIONS (Q1 – Q8)

Q1. Definition & Objective of Machine Learning

Definition: Machine Learning (ML) is a branch of AI enabling systems to learn patterns from historical data and improve performance automatically without explicit programming.

Primary Objective: Generalize from training data to accurately predict outcomes on unseen test data.

Real-World Example: *Email Spam Filter* — Learns word frequencies (e.g., "discount", "winner") from labeled emails to route new incoming messages to Spam or Inbox automatically.

Q2. Learning Paradigms & Applications

- **Supervised Learning:** Learns mapping from labeled pairs (X, Y). *App:* Housing price prediction.
- **Unsupervised Learning:** Discovers underlying structures in unlabeled data (X). *App:* Customer segmentation.
- **Reinforcement Learning:** Agent learns via trial/error actions to maximize cumulative rewards. *App:* Autonomous driving.

Q3. Machine Learning Life Cycle Phases

1. **Problem Formulation:** Define business target. 2. **Data Collection:** Gather raw inputs. 3. **Data Cleaning:** Impute missing values & remove outliers. 4. **EDA:** Map distributions & feature correlations. 5. **Feature Engineering:** Select/scale features. 6. **Model Training:** Fit parameters (weights/biases). 7. **Evaluation:** Assess accuracy metrics on unseen data. 8. **Deployment & Monitoring:** Serve predictions & track drift.

Q4. Dataset Splitting: Training vs Validation vs Testing

- **Training Set:** Direct fitting of model parameters (weights, biases).
- **Validation Set:** Hyperparameter tuning (e.g., tree depth, learning rate) and preventing overfitting.
- **Testing Set:** Completely unexposed data reserved for final unbiased performance evaluation.

Q5. [Numerical] Classification Accuracy

Given: Correct Predictions = 87, Total Instances = 120.

```
Accuracy = Correct Predictions / Total Instances = 87 / 120 = 0.725 = 72.5%
```

Q6. Training Accuracy 94% vs Testing Accuracy 69%

Identified Problem: **Overfitting (High Variance)**.

Justification: The model memorized noise and training anomalies rather than learning generalizable patterns, causing severe performance degradation on unseen test data.

Q7. Simple Linear Regression Model (Y = a + bX)

Slope (b): Measures the expected unit change in dependent variable Y for every 1-unit increase in independent variable X.

Intercept (a): Represents the baseline expected value of Y when X = 0.

Q8. Sigmoid Function in Logistic Regression

Maps real-valued linear outputs $z = w^T X + b$ to bounded range (0, 1):

SECTION 1: UNIT & CONCEPTUAL QUESTIONS (Q9 – Q16)

Q9. Naïve Independence Assumption

Assumption: Features are conditionally independent given the class label: $P(X_1, X_2|Y) = P(X_1|Y) \cdot P(X_2|Y)$.

Significance: Converts complex high-dimensional joint probability computations into simple product operations, eliminating data sparsity issues.

Q10. Information Gain & Decision Tree Splitting

Information Gain measures entropy reduction post-split: $IG(S, A) = Entropy(S) - \sum (|S_v| / |S|) Entropy(S_v)$.

Trees greedily select features yielding maximum IG to build homogenous nodes.

Q11. [Numerical] Performance Metrics Calculation

Given: TP = 52, TN = 47, FP = 13, FN = 8 (Total N = 120).

```
Accuracy = (TP + TN) / N = (52 + 47) / 120 = 99 / 120 = 82.50%
Precision = TP / (TP + FP) = 52 / (52 + 13) = 52 / 65 = 80.00%
Recall = TP / (TP + FN) = 52 / (52 + 8) = 52 / 60 = 86.67%
```

Q12. [Numerical] Regression Equation Derivation

Points: (2,3), (4,7), (6,11). n = 3, $\sum X = 12$, $\bar{x} = 4$, $\sum Y = 21$, $\bar{y} = 7$, $\sum X^2 = 56$, $\sum XY = 100$.

```
b = [n(∑XY) − (∑X)(∑Y)] / [n(∑X²) − (∑X)²] = [3(100) − (12)(21)] / [3(56) − 144] = 48 / 24 = 2.0
a = ȳ − b(x̄) = 7 − 2(4) = −1.0  ⇒  Y = −1 + 2X
```

Q13. Objectives & 4 Applications of ML

Objectives: Pattern discovery, regression, automated decision-making.

Applications: (1) *Healthcare:* Tumor segmentation; (2) *Finance:* Fraud scoring; (3) *E-Commerce:* Recommendations; (4) *Robotics:* Path planning.

Q14. Fundamental Principles of 5 Algorithms

- **ANN:** Multi-layer node networks processing weighted inputs via non-linear activation functions.
- **Decision Trees:** Recursive hierarchical feature space splitting based on purity criteria.
- **Bayesian Networks:** Directed Acyclic Graphs (DAGs) capturing probabilistic conditional dependencies.
- **SVM:** Max-margin hyperplanes separating classes in projected feature spaces.
- **Clustering:** Distance-metric grouping of unlabeled instances into cohesive clusters.

Q15. Data Science vs Machine Learning

Dimension	Data Science	Machine Learning
Objective	Extract actionable domain insights	Build automated predictive models
Data Scope	Wrangling, ETL, EDA, Visualization	Feature engineering, model tuning
Techniques	Statistics, SQL, Dashboards	Supervised/Unsupervised Algorithms

Q16. Machine Learning vs Deep Learning

ML: Requires explicit manual feature engineering; suitable for smaller tabular data (e.g., XGBoost on credit data).

DL: Hierarchical representation learning via multi-layer neural networks; scales with massive raw data (e.g.,

SECTION 1: CONCEPTUAL QUESTIONS (Q17 – Q24)

Q17. [Numerical] Comprehensive Binary Classification Metrics

Given: TP = 72, TN = 84, FP = 16, FN = 28 (Total N = 200).

```
Accuracy = (72 + 84) / 200 = 156 / 200 = 78.00%
Precision = 72 / (72 + 16) = 72 / 88 = 81.82%
Recall = 72 / (72 + 28) = 72 / 100 = 72.00%
F1-Score = 2 · (0.8182 · 0.72) / (0.8182 + 0.72) = 76.59%
```

Q18. Analysis of Model A vs Model B

Model A (Train 92%, Val 88%) exhibits strong generalization. **Model B (Train 98%, Val 67%) suffers from Overfitting** due to high performance degradation (31% drop) on validation data.

Q19. [Numerical] Fitness Workout Calories Regression

Data: X=[1,3,5,7,9], Y=[150,285,390,525,645]. n=5, ∑X=25, $\bar{x}$=5, ∑Y=1995, $\bar{y}$=399, $\sum X^2$=165, ∑XY=12435.

```
b = [5(12435) − (25)(1995)] / [5(165) − 625] = 12300 / 200 = 61.5
a = 399 − 61.5(5) = 91.5  ⇒  Y = 91.5 + 61.5X
Prediction (X = 11): Y = 91.5 + 61.5(11) = 768 Calories.
```

Suitability Comment: Extrapolating linearly to 11 hours ignores physiological exhaustion limits.

Q20. Ridge vs Lasso Regularization

Ridge (L2): Adds penalty $\lambda\sum w_i^2$. Shrinks weights close to zero; preserves all features.

Lasso (L1): Adds penalty $\lambda\sum|w_i|$. Forces coefficients to exact zero; performs sparse feature selection.

Q21. Decision Tree Construction Procedure

1. Compute baseline dataset entropy H(S). 2. For each attribute A, compute conditional entropy H(S|A). 3. Calculate Information Gain IG(S, A) = H(S) − H(S|A). 4. Select attribute with maximum IG as split node. 5. Recurse until leaf nodes reach complete purity or max depth.

Q22. SVM & Maximum-Margin Hyperplane

SVM constructs decision boundary $w^Tx + b = 0$. Maximizing functional margin $2/||w||$ between closest support vectors optimizes theoretical generalization bounds.

Q23. [Numerical] Least-Squares Regression Line

X={2,3,5,7}, Y={4,7,9,13}. n=4, ∑X=17, $\bar{x}$=4.25, ∑Y=33, $\bar{y}$=8.25, $\sum X^2$=87, ∑XY=165.

```
b = [4(165) − (17)(33)] / [4(87) − 289] = 99 / 59 = 1.678
a = 8.25 − 1.678(4.25) = 1.1185  ⇒  Y = 1.1185 + 1.678X
```

Q24. [Numerical] Naïve Bayes Spam Detection

Emails=12 (7 Spam, 5 Non-Spam). P(Discount|Spam)=5/7, P(Discount|Non-Spam)=2/5.

```
P(Discount) = (5/7)(7/12) + (2/5)(5/12) = 5/12 + 2/12 = 7/12
P(Spam|Discount) = [(5/7)(7/12)] / (7/12) = (5/12) / (7/12) = 5/7 ≈ 71.43%
```

SECTION 2: APPLICATION-BASED SCENARIOS (Q25 – Q37)

Q25. Student Exam Qualification Predictor

(a) Train: historical term records; Val: tuning depth; Test: current semester pool. (b) Life Cycle: Raw scores → Data cleaning → Model training → Validation → Deployment. (c) Metrics: F1-score & Accuracy. (d) Overfitting cause: Unconstrained tree depth on small class size.

Q26. Paradigm Classification & Feedback Types

- **Rainfall System:** Supervised (Feedback: Explicit continuous ground-truth measurements).
- **Music Grouping:** Unsupervised (Feedback: None; cluster compactness metrics).
- **Delivery Robot:** Reinforcement Learning (Feedback: Dynamic reward/penalty signals).

Q27. University ML Systems Analysis

Attendance: Supervised Classification | Logs → Risk Flag | Challenge: Sensor failure.
Library: Collaborative Filtering | Checkout History → Recommendations | Challenge: Cold start.
Exam Prediction: Regression | Test Marks → Final Grade | Challenge: Student motivational shifts.
Suspicious Login: Anomaly Detection | IP/Timestamp → Alert | Challenge: High class imbalance.

Q28 & Q29. Retail Recommendation Model Evaluation

Model A (Train 92%, Test 90%) generalizes cleanly. **Model B** (Train 97%, Test 73%) overfits. *Mitigation:* Apply L2 regularization, increase data volume, or prune features.

Q30. [Numerical] Academic Assistance Confusion Matrix

Actual Needing Help = 150 (Pos), Actual Not Needing = 450 (Neg). TP=120, FP=55, FN=30, TN=395.

```
Accuracy = (120+395)/600 = 85.83% | Precision = 120/175 = 68.57% | Recall = 120/150 = 80.00%
```

Key Metric: **Recall** is vital because failing to identify needy students (FN) harms student retention.

Q31. Hospital Clinical Patient Risk Model

Requires balanced clinical data, proper high-risk concept representation, robust non-linear function approximation, and safeguards against sensor noise and demographic sampling bias.

Q32. [Numerical] Real-Estate Price Prediction

Data: (4,36), (6,49), (8,61), (10,74). n=4, $\sum X=28$, $\bar{x}=7$, $\sum Y=220$, $\bar{y}=55$, $\sum X^2=216$, $\sum XY=1666$.

```
b = [4(1666) − (28)(220)] / [4(216) − 784] = 504 / 80 = 6.3 | a = 55 − 6.3(7) = 10.9
 ⇒  Y = 10.9 + 6.3X
```

Slope 6.3 indicates price rises $6.3k per 100 sq.ft; intercept 10.9 is base property value.

Q33–Q37. Additional Applied Model Analyses

Q33 Loan Default: Logistic regression outputs default probability using income/credit metrics.
Q34 Placement Tree: Split on attendance/GPA; prune weak branches to eliminate over-fitting.
Q35 Email Filter: Naïve Bayes calculates keyword likelihoods; assumes independence.
Q36 SVM Product Classifier: Maximizes margin between price/rating classes using kernel tricks.
Q37 Model Comparison: Model A (84%/81%) best generalized; Model B overfits; Model C has high bias.

SECTION 3: DETAILED NUMERICAL DERIVATIONS (Q38 – Q45)

Q38. [10 Marks] Logistic Regression Calculation

Model: $z = -4 + 0.7X_1 + 0.5X_2$. Given: $X_1 = 6$, $X_2 = 4$.

```
z = −4 + 0.7(6) + 0.5(4) = −4 + 4.2 + 2.0 = 2.2
σ(z) = 1 / (1 + e^(−2.2)) = 1 / (1 + 0.1108) = 0.9003
Threshold 0.5 Decision: 0.9003 ≥ 0.5  ⇒  Predicted Class = 1
```

Q39. [10 Marks] Logistic Regression Evaluation

Model: $z = -3 + 0.8X_1 + 0.6X_2$. Given: $X_1 = 5$, $X_2 = 3$.

```
z = −3 + 0.8(5) + 0.6(3) = −3 + 4.0 + 1.8 = 2.8
σ(2.8) = 1 / (1 + e^(−2.8)) = 1 / (1 + 0.0608) = 0.9426  ⇒  Predicted Class = 1
```

Q40. [10 Marks] Simple Linear Regression — Test Scores

Data: (2,48), (3,56), (5,67), (6,74), (8,85). n=5, ∑X=24, x̄=4.8, ∑Y=330, ȳ=66, $\sum X^2$=138, ∑XY=1723.

```
b = [5(1723) − (24)(330)] / [5(138) − 576] = [8615 − 7920] / [690 − 576] = 695 / 114
= 6.0965
a = 66 − 6.0965(4.8) = 36.7368  ⇒  Regression Equation: Y = 36.7368 + 6.0965X
Prediction for X = 7 hrs: Y = 36.7368 + 6.0965(7) = 79.41 Marks
```

Q41. [10 Marks] Least-Squares Regression — Assessment Scores

Data: (1,42), (3,54), (5,69), (7,78), (9,91). n=5, ∑X=25, x̄=5, ∑Y=334, ȳ=66.8, $\sum X^2$=165, ∑XY=1914.

```
b = [5(1914) − (25)(334)] / [5(165) − 625] = 1220 / 200 = 6.1
a = 66.8 − 6.1(5) = 36.3  ⇒  Regression Equation: Y = 36.3 + 6.1X
Prediction for X = 6 hrs: Y = 36.3 + 6.1(6) = 72.90 Marks
```

Q42 & Q43. [10 Marks] Multiple Linear Regression

Q42 Crop Yield: $Y = 18 + 2.5X_1 + 1.8X_2$. For X_1=6, X_2=5:

```
Y = 18 + 2.5(6) + 1.8(5) = 18 + 15 + 9 = 42 Yield Units
```

Q43 Sales Model: $Y = 25 + 3.2X_1 + 2.4X_2$. For X_1=7, X_2=5: Y = 25 + 22.4 + 12 = **59.4 Sales Units**. Coefficient 3.2 means sales rise by 3.2 units per unit increase in X_1.

Q44 & Q45. [10 Marks] Support Vector Machines

Q44 Point Classification: Decision Function $f(X) = 2X_1 + X_2 - 5$. Evaluated at new point (3,2):

```
f(3,2) = 2(3) + 2 − 5 = +3  ⇒  Since f(X) > 0, Class = +1
```

Q45 SVM Hyperplane: Separates Class +1 [(2,3),(3,5)] from Class -1 [(6,1),(7,2)]. Support vectors lie on the dynamic margin boundaries $w^Tx + b = \pm 1$.

SECTION 4: ADVANCED METRICS & LDA SOLUTIONS (Q46 – Q51)

Q46. [10 Marks] Linear Discriminant Analysis — Class Means

Class 1: (2,4), (4,6), (6,5). Class 2: (8,3), (7,1), (9,2).

$$\text{Class 1 Mean } \mu_1 = ((2+4+6)/3,\ (4+6+5)/3) = (4.0,\ 5.0)$$
$$\text{Class 2 Mean } \mu_2 = ((8+7+9)/3,\ (3+1+2)/3) = (8.0,\ 2.0)$$
$$\text{Overall Mean } \mu = ((12+24)/6,\ (15+6)/6) = (6.0,\ 3.5)$$

Q47. [10 Marks] LDA Classification by Distance

Class 1: (2,3), (3,5), (4,4) ⇒ $\mu_1 = (3, 4)$. Class 2: (7,2), (8,4), (9,3) ⇒ $\mu_2 = (8, 3)$. Point X = (5, 4).

$$d(X, \mu_1) = \sqrt{[(5-3)^2 + (4-4)^2]} = \sqrt{4} = 2.00$$
$$d(X, \mu_2) = \sqrt{[(5-8)^2 + (4-3)^2]} = \sqrt{(9 + 1)} = \sqrt{10} \approx 3.16$$
$$\text{Since } d_1 < d_2 \ (2.00 < 3.16) \ \Rightarrow \ \textbf{Assign X to Class 1}$$

Q48 & Q49. [10 Marks] Naïve Bayes Classification Procedure

Calculate class priors $P(C_j)$ and conditional likelihoods $P(X_i | C_j)$. For target instance X*:

$$\text{Score}(C_j) = P(C_j) \cdot \prod P(X_i | C_j)$$
$$\text{Assign Class} = \text{argmax}_{C_j} [\ \text{Score}(C_j)\ ]$$

Q50. [10 Marks] Full Model Evaluation Metrics

Given: TP = 48, TN = 44, FP = 12, FN = 16 (Total N = 120).

$$\text{Accuracy} = (48 + 44) / 120 = 92 / 120 = 76.67\%$$
$$\text{Precision} = 48 / (48 + 12) = 48 / 60 = 80.00\%$$
$$\text{Recall} = 48 / (48 + 16) = 48 / 64 = 75.00\%$$
$$\text{F1-Score} = 2 \cdot (0.80 \cdot 0.75) / (0.80 + 0.75) = 1.20 / 1.55 = 77.42\%$$

Interpretation: 80% of positive predictions were true positive alerts; 75% of actual positive instances were successfully identified by the system.

Q51. [10 Marks] Regression Error Metrics (MSE, RMSE, R^2)

Actual Y = [62, 71, 78, 85, 92], Predicted $\hat{Y}$ = [60, 74, 75, 88, 89]. Mean $\bar{y} = 77.6$.

Residuals ($Y - \hat{Y}$): [2, −3, 3, −3, 3] ⇒ Squared Errors: [4, 9, 9, 9, 9]. Sum RSS = 40.

$$MSE = RSS / n = 40 / 5 = 8.00$$
$$RMSE = \sqrt{8.00} \approx 2.83 \text{ Marks}$$
$$TSS = \sum(Y - \bar{y})^2 = (-15.6)^2 + (-6.6)^2 + (0.4)^2 + (7.4)^2 + (14.4)^2 = 549.20$$
$$R^2 = 1 - (RSS / TSS) = 1 - (40 / 549.20) = 1 - 0.0728 = 0.9272\ (92.72\%)$$

Interpretation: An R^2 of 92.72% demonstrates an exceptionally high proportion of variance explained by the model, paired with a low average prediction error of 2.83 marks (RMSE).